A Design Procedure for Aperture Coupled Microstrip Antennas Based on Approximate Equivalent Networks

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Abstract

A design procedure for aperture coupled microstrip antennas based on approximate equivalent networks is presented. This is then used in numerical simulations as an initial-starting geometry in order to get an accurate design. Approximate expressions for the effective dielectric constant and the natural resonance frequency of the patch and the slot are given. These are in turn related to the resonant frequency of the whole antenna through simplified equivalent networks, which is then made identical to the operating frequency. A design example is carried out and verified through numerical simulations.

I. Introduction

The aperture coupled patch antennas have found widespread applications in communication systems during the last decade. They are also expected to be employed in radar systems, especially in the form of conformal arrays. Their most important advantage that makes them preferable from microstrip line or probe fed patches is the ability to use separate substrates for the feeding network and the patch itself. In their usual structure a patch printed on a low dielectric constant substrate (e.g. foam) is excited by an aperture in the ground plane which is in turn excited by a microstrip line printed on a separate substrate covering the other side of the ground plane. This configuration reduces the surface wave effects on the patch antenna, it enhances its bandwidth, it is isolating the feeding lines spurious radiation and leaves more space for the feeding network. This latter advantage makes them very attractive for phased arrays applications where phase shifters and/or power dividers must be incorporated in the feeding network. Moreover, modern frequency agile radars as well as electronic warfare applications require printed antennas with very large bandwidth, but these should also be of conformal nature. The aperture coupled antennas could serve this need very well provided that a repeatable and relatively simple design procedure is first established for both planar and at least cylindrical substrate geometries. The present work is restricted to the planar geometry. The work on conformal aperture coupled antennas is quite limited and there are a lot of subjects that should be analyzed first, before moving to their design.

Recently, a modification is introduced by a number of authors, e.g. [1]; namely a double layer substrate is used under the patch. The layer directly covering the ground plane has a higher dielectric constant (e.g. duroid, $\varepsilon_r \approx 2$) and is used to print a second feeding network on it. The purpose of this modification is to use two uncoupled apertures each one excited by a different feed line and each aperture provides in turn a separate linear polarization of the patch. In this manner dual linear polarization or circular polarization arrays can be designed. In both of the above geometries a foam layer is used below the patch, thus a superstrate is needed in order to print the patch on its lower side. Also a foam layer and a backplane is needed on the lower side (opposite of the patch) to prevent the slot back-radiation and to make the structure unidirectional, [2-3].

It is obvious from the above that this structure is a rather complicated multilayer one. In most of the cases a purely numerical design approach is employed, but this may lead to an excessive computer and man-time effort. What is needed, which is also the purpose of the present work, is a good approximate starting solution. An equivalent circuit approach is employed for this purpose. Recall first that in probe-fed patch elements and at a very good approximation in microstrip line fed ones, the patch natural frequency coincides with the resonant frequency of the whole structure. This is not the case in aperture coupled patches. The structure input impedance as seen by the feeding line consists of two electromagnetically coupled equivalent circuits for the patch and the aperture. Realizing that the antenna operating frequency should coincide with the resonant frequency of the whole structure (f_{tr}), this work aims at expressions relating the patch natural resonant frequency (f_{pr}) and the aperture inductance (or again its natural resonant frequency) to f_{tr}. In turn these values are used to estimate the patch and aperture dimensions through convenient formulas.

II. Formulation

The aperture coupled microstrip antenna in its simpler form originally proposed by Pozar [4], is shown in Fig.1. The antenna excitation, or equivalently the coupling mechanism between the microstrip line and the slot and between the slot and the patch, was investigated in [4]. The two coupling mechanisms are very similar, since the patch can be considered as an open-ended wide microstrip line. For a maximum in the coupling between the slot and the patch it was found in [4] that the slot should be rectangular and positioned parallel to the two radiating edges along their mid-distance (e.g. along x=0 in Fig. 1), [5, 6]. This configuration can be considered as an excitation of the TM_{100} mode, while a slot orthogonal to this one (parallel to the other two edges) excites the TM_{010} mode.

Concerning the coupling between the microstrip line and the aperture, it was found in [7] that the slot should be perpendicular to the feedline and its width (w_s) should be small enough to be considered a small perturbation to the feedline (quasi-TEM) fields. Moreover, the slot should be long enough to avoid interactions between the feedline field and the slot edges according to [8, p.16] and [7], who have also defined a lower limit as:

$$\ell_{slot} > w_{line} + n \cdot d_{sub} \qquad \text{with } n > 6 \tag{1}$$

where w_{line} = feedline width and d_{sub} its substrate thickness.

Moreover, Pozar [9] recommends a rule of thumb in choosing the ratio of slot width over its length as :

$$w_{slot} / \ell_{slot} = 1/10 \tag{2}$$

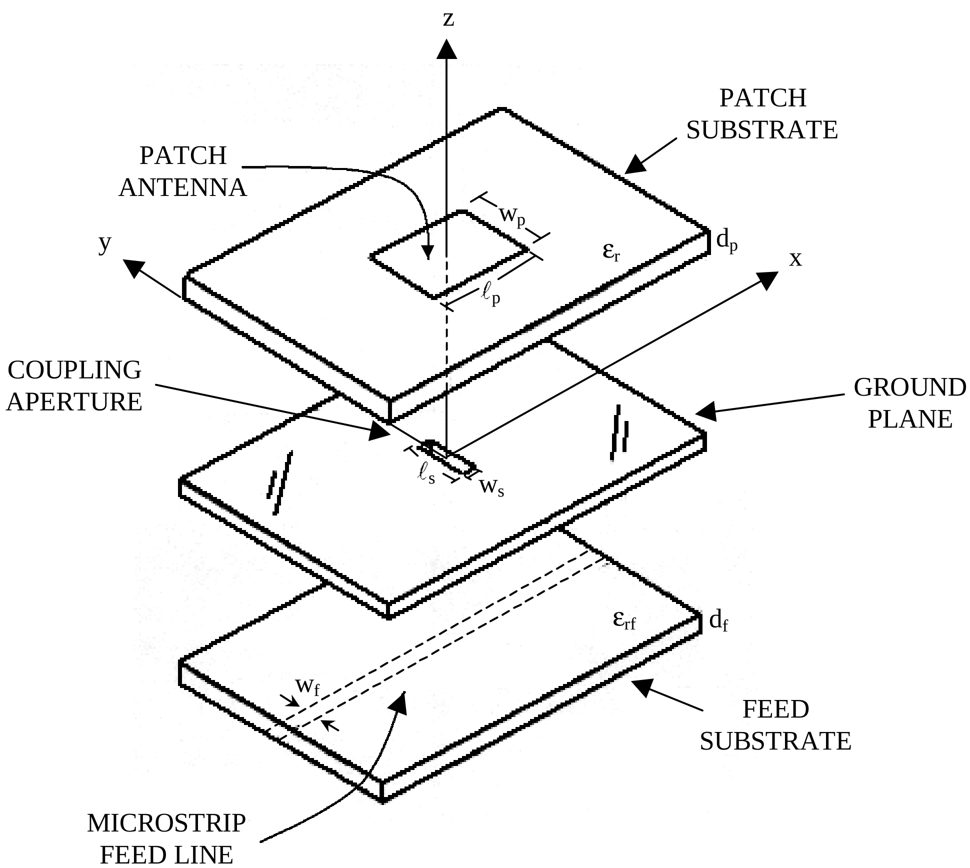


Fig. 1. The original single feed aperture coupled microstrip antenna proposed by Pozar [4].

As shown in Fig. 1 the excitation of the single aperture can be done with a single microstrip line transverse to the slot and running through its center. However, when the two orthogonal apertures are employed the excitation of each one of them must be done with a "balanced pair" of microstrip lines "symmetrically located" about the center of the slot, [10]. In this manner the two apertures are well decoupled, avoiding cross-polarization effects in the antenna radiation pattern. The term "balanced pair" means that the two lines must be fed by signals of equal amplitude and phase. The "fork-feed" proposed by Yamazaki et al [11] in turn seems to be a very good idea for achieving accurate balance, since the two arms have a common input-port and they will be made exactly symmetrical. It is assumed that the two orthogonal apertures used in dual-linear and circular polarization are excited by two "fork-feeds" printed on substrates above and below the ground plane (on which the slots are etched). The fact that the two orthogonal apertures are very well decoupled is important, since each one of the two feeding networks (upper and lower) can be treated separately. Namely, the following analysis can be restricted to the single aperture of Fig.1.As noted above, the patch and slot natural resonant frequencies are different from that of the whole antenna, which is in turn identical to the operating frequency. The present design approach is based on two basic steps:

i) Find expressions relating the patch and slot natural resonant frequencies to their dimensions and the characteristics of the multilayer structure.
ii) Find expressions or a procedure relating the whole antenna resonant frequency to the patch and slot own natural resonant frequencies.

Starting from the aperture characteristics, a fairly good and simple approximation is to consider the slot as a rectangular waveguide section oriented in the z-direction. At the resonance the waveguide (slot) can be considered as a cavity ending upon "magnetic walls" on the top and bottom layers. Usually the metallic slot layer is thin, so the dominant resonant mode will be the TE_{100}. Also there is actually a leakage through these walls, then in order to calculate the resonant frequency we could consider an equivalent cavity with perfect magnetic walls but filled with a dielectric with ε_{req}. In [2, 6] they used the expression:

$$\varepsilon_{req} \approx (\varepsilon_{r1} + \varepsilon_{r2}) / 2 \tag{3}$$

while a different expression for ε_{req} is proposed in [12] as :

$$\varepsilon_{req} = \frac{2\varepsilon_{r1}\varepsilon_{r2}}{\varepsilon_{r1} + \varepsilon_{r2}} \tag{4}$$

Alternatively, we propose herein the modeling and analysis of a slotline with the same width as the one which is to be used for the patch excitation. This can be easily done using the moment method program included in HP-MDS software (called Momentum) or any other MOM-technique. The whole multilayer structure is modeled but only a slotline (aperture) with two ports at its small sides is assumed. Then, from the results for the S_{21} the propagation constant ($\gamma=\alpha+j\beta$) is estimated. Since $\beta = k_0\sqrt{\varepsilon_{req}}$ then simply take :

$$\varepsilon_{req} = (\beta/k_0)^2 \tag{5}$$

Thus, considering the slot as a cavity resonating at the TE_{100} mode and using one of the Eqs. (3)-(5) to find ε_{req}, then its resonant frequency can be calculated as:

$$f_{sr} = \frac{c}{2\ell_s\sqrt{\varepsilon_{req}}} \tag{6}$$

Our next task is to estimate the patch natural resonant frequency. The modes of interest are the TM_{100} and TM_{010}. The cavity model approximation can be employed again assuming "magnetic walls" around the edges of the patch, but filled with an equivalent dielectric ε_{reff} (accounting for the leakage through these walls). The approximate resonant frequency is then [13, p.10] :

$$TM_{pq0} \text{ mode : } f_{pr}^{p,q} = \frac{c}{2\sqrt{\varepsilon_{reff}}}\sqrt{\left(\frac{p}{\ell_p}\right)^2 + \left(\frac{q}{w_p}\right)^2} \tag{7}$$

When dual linear polarization is desired the resonant frequencies of TM_{100} and TM_{010} should be identical and this can be done assuming a square patch ($\ell_p=w_p$). In this manner, the problem of estimating the patch resonant dimensions is reduced to the estimation of its effective dielectric constant.

For wide microstrip lines (w/d >> 1), as it is the case for the microstrip line with width equal to that of the patch, the ε_{reff} is not a critical function of their normalized width (w/d). Thus, an approximate value for the width w can be considered. For a very rough approximation consider an equivalent homogeneous substrate with a thickness $d_t = \sum d_i$ and dielectric constant $\varepsilon_{rav} = \left(\sum d_i\varepsilon_{ri}\right)/d_t$ where d_i, ε_{ri} the thickness and the dielectric constant of the layers between the ground plane (slot layer) and the patch (ignoring the presence of a superstrate). Calculate then the wavelength for a TEM mode in an unbounded media with $\varepsilon_r=\varepsilon_{rav}$ as $\lambda_u = c/2\sqrt{\varepsilon_{rav}}$, so that a rough approximation of width (w_{pu}) can be taken between $\lambda_u/2$ and $\lambda_u/4$ and preferably toward the $\lambda_u/2$. For a better approximation the values of ε_{rav}, d_t and w_p are used in the usual expression of ε_{reff}, for wide microstrip lines, e.g. [13, p.10].

For a more accurate estimation of the effective dielectric constant a multilayer model of a microstrip line should be used. Alternatively and more practically is to use the HP-MDS software. Either the moment method (momentum) or its multilayer options can be used. In both cases all the layers above the slot layer (acting as ground plane) are modeled, including the superstrate and any adhesive layers employed for bonding purposes. A microstrip line of width w_p and arbitrary length ℓ_p but preferably greater than w_p, is considered at the patch position. A simulation in the desired frequency range is then carried out in a similar manner as for the slot-line and the complex propagation constant $\gamma=\alpha+j\beta$ is then evaluated. The effective dielectric constant is again calculated as $\varepsilon_{reff}=(\beta/k_0)^2$. The multilayer models included in HP-MDS (or HP-ADS) are very fast and are obviously preferable and the only doubt about them is to prove that they are equally accurate as the Momentum solution.

As pointed out in [7] and observed by all the other investigators of this field, the slot as well as the whole antenna input impedance appears as a series element (Z_{series}) to the microstrip feeding line. In both simulations and measurements the impedance (or reflection coefficient) at the input-port of the feeding network is obtained. But for both design and comparison purposes the required quantity is the antenna input impedance as seen by the feeding line (Z_{series}) with the reference plane passing through the slot center. For this reason a de-embedding technique is needed, which is already given in our previous works, [5, 6].

The problem that remains is to relate the whole structure resonant frequency (which should be identical to the operating frequency) with the above given natural resonances. This can be done through an equivalent circuit. This consists of a series resonant circuit for the patch and a parallel resonant circuit for the slot as shown in Fig.2. The electromagnetic coupling of the patch-slot and slot-feedline is presented by transformers.

The slot acts as series impedance seen from the feeding line, [2, 14, 15] and it has almost pure inductive behaviour when operated below resonance while it can be represented by a parallel resonant circuit in general. The presence of a patch over the slot reacts as admittance in parallel with the slot equivalent circuit, [14, 15]. It is very important to note here that *the patch behaves as "a series resonant circuit"* in contrast to the usually assumed parallel resonance behavior. This is in fact a very well explained principle, since Harrington (1961), [16, p.434], clearly clarifies this phenomenon. Thus, a probe fed cavity or equivalently a cavity excited by an electric current behaves as a parallel resonant circuit. In contrast, an aperture coupled cavity or equivalently a cavity excited by a magnetic current behaves as a series resonant circuit. But, the whole antenna (combined patch+slot) behaves as a parallel resonant circuit. The resonances occur actually at the frequency where the patch and slot reactive parts are equal but with opposite signs, namely where the slot inductive part compensates the patch capacitive part. Considering a non-resonant slot, its inductive behaviour lowers the resonant frequency of the aperture coupled antenna with respect to the patch natural resonant frequency as it is already proved in [15].

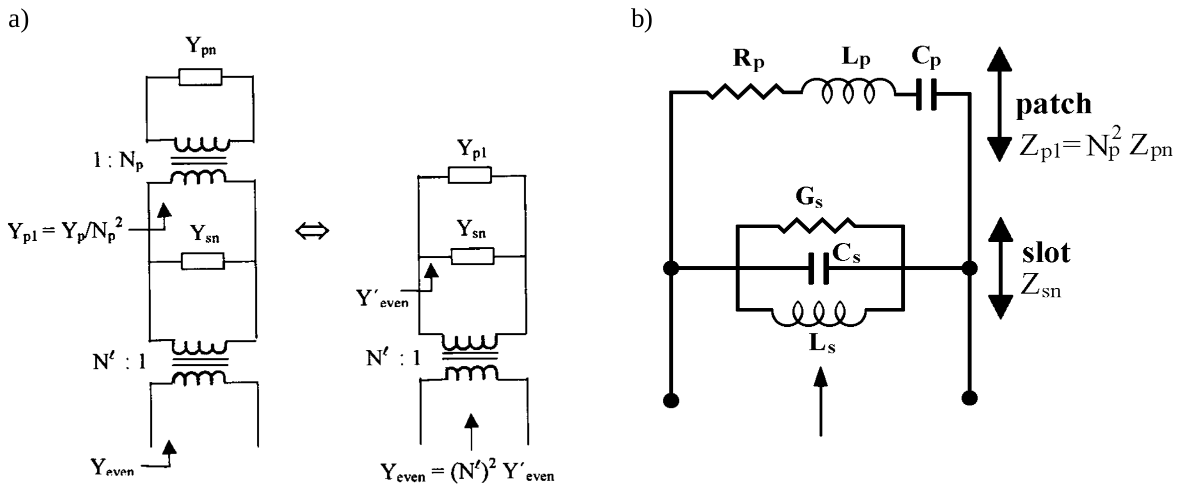


Fig.2 a) Equivalent circuit of the aperture coupled patch antenna as seen by the feeding-line with transformers representing the mutual couplings. Y_{pn} and Y_{sn} represent the patch and slot self-admittances.

b) The equivalent circuit for the dominant mode at the slot level.

The basic idea for the design is the following : i) Estimate an approximate value for the patch and slot equivalent circuit components (Fig.2b). ii) Find the resonant frequency of the whole structure by analyzing or simulating the circuit of Fig.2b and iii) Adjust-tune the circuit components so that the whole circuit resonant frequency becomes equal the desired operating frequency. In turn, the resulting values give the proper patch and slot natural resonant frequencies from which their dimensions are estimated through Eqs. (6) and (7).

A simplified approach is to use a "typical" resonance curve for the patch impedance and the slot admittance, for the specific multilayer structure on which the patch is printed. It is very useful to recall here that for a parallel resonant circuit (like that of the slot), we only need its resonant frequency (f_{res}=f_{rs}) and its admittance value $Y_s(f)$ at only one arbitrary frequency (preferably within its bandwidth) to estimate its equivalent capacitance (C_s) and inductance (L_s), as :

$$\text{Slot:} \quad L_s = \frac{(f/f_{res})^2 - 1}{2\pi f \ \mathrm{Im}\{Y_s(f)\}}, \qquad C_s = 1/\{L_s(2\pi f_{res})^2\}, \quad G_s = \mathrm{Re}\{Y_s(f_{res})\} \tag{8}$$

Similar expressions can be easily deduced for the patch series resonant circuit.

$$\text{patch :} \quad C_p = \frac{(f/f_{res})^2 - 1}{2\pi f \ \mathrm{Im}\{Z_p(f)\}}, \qquad L_p = 1/\{C_p(2\pi f_{res})^2\}, \quad R_p = \mathrm{Re}\{Z_p(f_{res})\} \tag{9}$$

where f_{res}= f_{rp} is the patch resonant frequency and $Z_p(f)$ its impedance at an arbitrary frequency f.

Also, the conductance of the slot can be estimated at its resonance according to [16, p.444] as:

$$\text{Resonant slot: } \ell_s = \lambda_s/2 \rightarrow G_s = \mathrm{Re}(Y_s) \approx 0.004(w_s/\lambda_s) = 0.002(w_s/\ell_s) \tag{10}$$

It is important to note that the patch and slot natural resonant frequencies are preserved through the coupling.

One way to find "typical" resonance curves for the patch and the slot is to simulate each one of them separately (e.g. with HP-MDS Momentum). On the other hand it may be preferable to use approximate models. For this purpose the patch impedance can be obtained from a "Transmission Line Model", e.g. [14]. Likewise, the slot is considered as two short-circuited slot lines [17]. However, for the slot it is preferable to simulate the structure in the absence of the patch in HP-Momentum.

The whole circuit presents multiple resonances and the choice depends just on which condition is preferable. For example, one may prefers to operate the antenna with the patch and slot resonances close to each other to enhance the bandwidth, or it may prefers to use a slot far below its resonance to avoid strong parallel plate modes excitation (when a back-plane is added in order to make the slot uni-directional).

III. Numerical Results

In order to verify the above given procedure a design of an aperture coupled microstrip antenna operating at 5.0GHz is considered. The numerical results given in [14] for a single feed, single patch substrate geometry as shown in Fig.1 are used as "typical resonant curves" for the patch and the slot. Its data are : Patch: ℓ_p=w_p=21.5mm and ε_{rp}=1.05, slot: ℓ_s=15mm, w_s=2mm and feedline substrate ε_{rf}=2.33. These give patch and slot resonances at f_{rp}=5.6GHz and f_{rs}=8.9GHz. While the corresponding equivalent circuit components are:

R_p=43.7Ω, C_p=0.08pF, L_p=10nH, G_s=0.2mS, C_s=0.125pF, L_s=2.56nH

The aim here is to design a multilayer aperture coupled antenna operating at 5.0GHz. The multilayer geometry includes adhesive layers and a superstrate (the patch is printed on its lower side) as well as a "fork-feed" for the slot excitation. The layers characteristics are as follows:

Feed substrate:	Duroid 5870, ε_r=2.33, tanγ=0.0012, thickness d_f=0.7874mm
Patch substrate:	Foam (Rohacell HF51), ε_r=1.1, tanγ=0.0018, thickness d_f=5mm
Patch superstrate:	Duroid 5870, ε_r=2.33, tanγ=0.0012, thickness d_f=0.3937mm

The previously described procedure is then applied in order to evaluate the patch and slot dimensions so that the whole antenna resonates at 5.0GHz. This yields:

patch: ℓ_p=w_p=18.9mm and natural resonance at f_{rp}=6.34GHz

slot: ℓ_s=18.4mm, w_s=2mm and natural resonance at f_{rs}=7.25GHz

With an equivalent circuit data:

R_p=43.7Ω, C_p=0.31pF, L_p=2nH, G_s=0.22mS, C_s=0.152pF, L_s=3.15nH

The equivalent circuit of the whole structure present two parallel resonances (maximum of the real part) at f_{rt}=4.94 and 9.36GHz as one can see from Fig. 3. Also, the real part of the input impedance appears to be very high (of the order of 650Ω). This is in fact very far away from the true one, since the method aims at the estimation of the correct operating frequency and it does not account for the impedance magnitude.

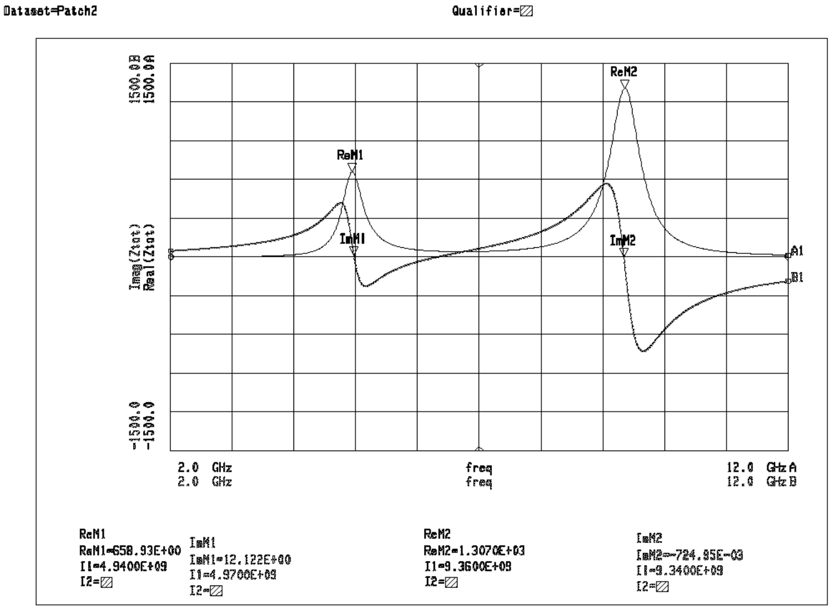
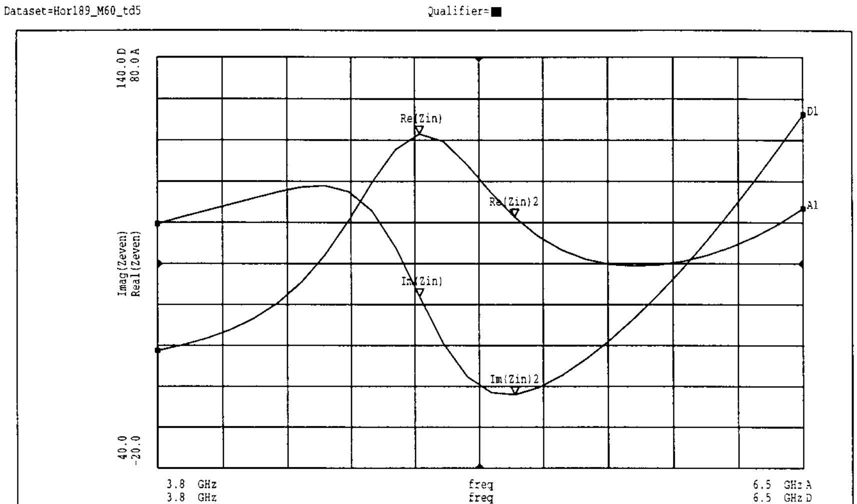


Fig. 3. The whole antenna input impedance (Z_{tot}=Z_{even}) at the slot level calculated from the equivalent circuit of Fig. 2.

A numerical simulation employing the HP-Momentum (Moment method) software is then applied in order to tune the solution. The microstrip lines (Z_0=50, 100Ω) involved in the "fork-feed" are separately characterized using both the HP-MDS and the HP-Momentum. The de-embedding described in [6] is in turn applied to get the input impedance at the slot level. The results for the input impedance are shown in Fig. 4.



DEEMBEDDING OF SIMULATED RESULTS

```
Omega1=2.*PI*freq
Move the Reference Plane to the Tee input
Ereff0=2.111
Alpha0=2.596E-3/8.686
L0=7
Beta0=Omega1*sqrt(Ereff0)/3.E11
Glexp0=exp(-j*2*Beta0*L0)*exp(-2*Alpha0*L0)
S11t=S11/Glexp0
```

```
Calculate the 100 ohms lines characteristics
Ereff1=2.072
Z02a=84.222
Alpha1a=0.491E-3
L1a=18.0
L2a=10.0
Lta=L1a+L2a
Beta1a=Omega1*sqrt(Ereff1)/3.E11
Gam1a=Alpha1a+j*Beta1a
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```
Calculate Zeven
SB=exp(-2*Gam1a*L1a)
SC=exp(-2*Gam1a*L2a)
SA=SB*SC
SST1=(S11t-SA)
SST2=(SC-1)*(S11t-SB)
Zeven=2*Z02a*SST1/SSt2
```

Fig.4. The antenna input impedance (Z_{even}) after de-embedding the results of an HP-Momentum simulation.

The characteristics as well as the equations used for the de-embedding appear at the bottom of Fig. 4. The input impedance presents a resonance at f_r=4.9GHz with a maximum real part equal to R_{max}=61.5Ω. The corresponding desired values are f_r=5.0GHz and R_{max}=100Ω. The patch and slot dimensions should then be tuned repeating the numerical simulations to get a resonance at 5.0GHz. Also an antenna design operating at 5.3GHz is under consideration. Results for both of them will be available at the time of the presentation.

Conclusions

A design procedure for aperture coupled microstrip antennas based on approximate equivalent networks and "typical" patch and slot resonance curves is presented. An example of an antenna design at 5.0GHz is also carried out. A necessary extension of this work should include the evaluation of the patch and slot equivalent networks from their transmission line models. But, more important is to work toward the design of aperture coupled antennas printed on cylindrical substrates.

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